P2 Chapter 9: Differentiation

Rates of Change

Relating Rates of Change

Q Determine the rate of change of the area A of a circle when the radius $r = 3$cm, given that the radius is changing at a rate of $5\ cm\ s^{-1}$.

Firstly, how would we represent…

"the rate of change of the area A" $\quad \boldsymbol{\frac{dA}{dt}}$

"the rate of change of the radius r is 5" $\quad \boldsymbol{\frac{dr}{dt} = 5}$

"the area A of a circle" $\quad \boldsymbol{A = \pi r^2}$

Tip: Whenever you see the word '<u>rate</u>', think $/dt$

Then by Chain Rule:

First copy top and bottom into the diagonals…

$$\frac{dA}{dt} = \frac{\quad}{dr} \times \frac{dr}{\quad}$$

…Then fill in the gaps with whatever variable you didn't use.

$$= 2\pi r \times 5$$
$$= 2\pi(3) \times 5$$
$$= 30\pi$$

Further Example

Edexcel C4 June 2008 Q3

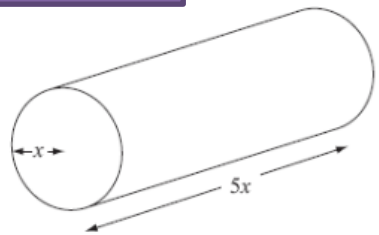

Figure 2 shows a right circular cylindrical metal rod which is expanding as it is heated. After t seconds the radius of the rod is x cm and the length of the rod is $5x$ cm.
The cross-sectional area of the rod is increasing at the constant rate of $0.032\ \text{cm}^2\,\text{s}^{-1}$.

(*a*) Find $\frac{dx}{dt}$ when the radius of the rod is 2 cm, giving your answer to 3 significant figures. **(4)**

(*b*) Find the rate of increase of the volume of the rod when $x = 2$. **(4)**

It's common to have a second part of the question where you have to connect further rates.

(*a*)

From question, $\frac{dA}{dt} = 0.032$	B1
$\left\{A = \pi x^2 \Rightarrow \frac{dA}{dx} =\right\} 2\pi x$	B1
$\frac{dx}{dt} = \frac{dA}{dt} \div \frac{dA}{dx} = (0.032)\frac{1}{2\pi x};\ \left\{= \frac{0.016}{\pi x}\right\}$	M1
When $x = 2\text{cm}$, $\frac{dx}{dt} = \frac{0.016}{2\pi}$	
Hence, $\frac{dx}{dt} = 0.002546479...\ (\text{cm s}^{-1})$	A1 cso (4)

(*b*)

$V = \underline{\pi x^2(5x)} = \underline{5\pi x^3}$	B1
$\frac{dV}{dx} = 15\pi x^2$	B1 ft
$\frac{dV}{dt} = \frac{dV}{dx} \times \frac{dx}{dt} = 15\pi x^2.\left(\frac{0.016}{\pi x}\right);\ \{= 0.24x\}$	M1
When $x = 2\text{cm}$, $\frac{dV}{dt} = 0.24(2) = \underline{0.48}\ (\text{cm}^3\,\text{s}^{-1})$	A1 (4)

Test Your Understanding

June 2012 Q2

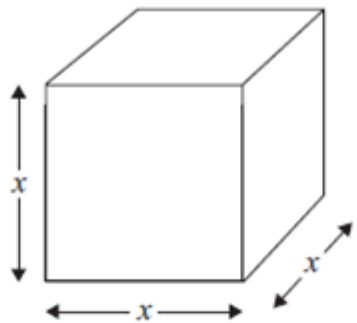

Figure 1

Figure 1 shows a metal cube which is expanding uniformly as it is heated.

At time t seconds, the length of each edge of the cube is x cm, and the volume of the cube is V cm^3.

(*a*) Show that $\frac{dV}{dx} = 3x^2$. **(1)**

$$V = x^3$$
$$\frac{dV}{dx} = 3x^2$$

Given that the volume, V cm^3, increases at a constant rate of 0.048 cm^3 s^{-1},

(*b*) find $\frac{dx}{dt}$ when $x = 8$, **(2)**

$$\frac{dx}{dt} = \frac{dx}{dV} \times \frac{dV}{dt} = \frac{1}{3 \times 8^2} \times 0.048$$
$$= 0.00025$$

(*c*) find the rate of increase of the total surface area of the cube, in cm^2 s^{-1}, when $x = 8$. **(3)**

Fro Tip: Don't forget that if you know $\frac{dV}{dx}$ then $\frac{dx}{dV} = \frac{1}{dV/dx}$

$$A = 6x^2$$
$$\frac{dA}{dx} = 12x$$
$$\frac{dA}{dt} = \frac{dA}{dx} \times \frac{dx}{dt}$$
$$= (12 \times 8) \times 0.00025 = 0.024$$

Homework Exercise

1 Given that $A = \frac{1}{4}\pi r^2$ and that $\frac{dr}{dt} = 6$, find $\frac{dA}{dt}$ when $r = 2$.

2 Given that $y = xe^x$ and that $\frac{dx}{dt} = 5$, find $\frac{dy}{dt}$ when $x = 2$.

3 Given that $r = 1 + 3\cos\theta$ and that $\frac{d\theta}{dt} = 3$, find $\frac{dr}{dt}$ when $\theta = \frac{\pi}{6}$

4 Given that $V = \frac{1}{3}\pi r^3$ and that $\frac{dV}{dt} = 8$, find $\frac{dr}{dt}$ when $r = 3$.

5 A population is growing at a rate which is proportional to the size of the population. Write down a differential equation for the growth of the population.

6 A curve C has equation $y = f(x)$, $y > 0$. At any point P on the curve, the gradient of C is proportional to the product of the x- and the y-coordinates of P. The point A with coordinates (4, 2) is on C and the gradient of C at A is $\frac{1}{2}$

Show that $\frac{dy}{dx} = \frac{xy}{16}$

7 Liquid is pouring into a container at a constant rate of $30\,\text{cm}^3\,\text{s}^{-1}$. At time t seconds liquid is leaking from the container at a rate of $\frac{2}{15}V\,\text{cm}^3\,\text{s}^{-1}$, where $V\,\text{cm}^3$ is the volume of the liquid in the container at that time.

Show that $-15\frac{dV}{dt} = 2V - 450$.

Homework Exercise

8 An electrically-charged body loses its charge, Q coulombs, at a rate, measured in coulombs per second, proportional to the charge Q.
Write down a differential equation in terms of Q and t where t is the time in seconds since the body started to lose its charge.

9 The ice on a pond has a thickness x mm at a time t hours after the start of freezing. The rate of increase of x is inversely proportional to the square of x.
Write down a differential equation in terms of x and t.

10 The radius of a circle is increasing at a constant rate of 0.4 cm per second.

a Find $\frac{dC}{dt}$, where C is the circumference of the circle, and interpret this value in the context of the model.

b Find the rate at which the area of the circle is increasing when the radius is 10 cm.

c Find the radius of the circle when its area is increasing at the rate of 20 cm^2 per second.

11 The volume of a cube is decreasing at a constant rate of 4.5 cm^3 per second. Find:

a the rate at which the length of one side of the cube is decreasing when the volume is 100 cm^3

b the volume of the cube when the length of one side is decreasing at the rate of 2 mm per second.

12 Fluid flows out of a cylindrical tank with constant cross section. At time t minutes, $t > 0$, the volume of fluid remaining in the tank is V m^3. The rate at which the fluid flows in $m^3\,min^{-1}$ is proportional to the square root of V.
Show that the depth, h metres, of fluid in the tank satisfies the differential equation $\frac{dh}{dt} = -k\sqrt{h}$, where k is a positive constant.

Homework Exercise

13 At time, t seconds, the surface area of a cube is $A\,\text{cm}^2$ and the volume is $V\,\text{cm}^3$.
The surface area of the cube is expanding at a constant rate of $2\,\text{cm}^2\,\text{s}^{-1}$.

a Write an expression for V in terms of A.

b Find an expression for $\frac{\mathrm{d}V}{\mathrm{d}A}$

c Show that $\frac{\mathrm{d}V}{\mathrm{d}t} = \frac{1}{2}V^{\frac{1}{3}}$

14 An inverted conical funnel is full of salt. The salt is allowed to leave by a small hole in the vertex. It leaves at a constant rate of $6\,\text{cm}^3\,\text{s}^{-1}$.
Given that the angle of the cone between the slanting edge and the vertical is 30°, show that the volume of the salt is $\frac{1}{9}\pi h^3$, where h is the height of salt at time t seconds. Show that the rate of change of the height of the salt in the funnel is inversely proportional to h^2. Write down a differential equation relating h and t.

Homework Answers

1 6π **2** $15e^2$ **3** $-\frac{9}{2}$ **4** $\frac{8}{9\pi}$ **5** $\frac{dP}{dt} = kP$

6 $\frac{dy}{dx} = kxy$; at (4, 2) $\frac{dy}{dx} = \frac{1}{2}$, so $8k = \frac{1}{2}$, $k = \frac{1}{16}$

Therefore $\frac{dy}{dx} = \frac{xy}{16}$

7 $\frac{dV}{dt}$ = rate in − rate out $= 30 - \frac{2}{15}V \Rightarrow 15\frac{dV}{dt} = 450 - 2V$

So $-15\frac{dV}{dt} = 2V - 450$

8 $\frac{dQ}{dt} = -kQ$ **9** $\frac{dx}{dt} = \frac{k}{x^2}$

10 **a** Circumference, C, $= 2\pi r$, so $\frac{dC}{dt} = 2\pi \times 0.4$

$= 0.8\pi \text{ cm s}^{-1}$

Rate of increase of circumference with respect to time.

b $8\pi \text{ cm}^2 \text{ s}^{-1}$ **c** $\frac{25}{\pi}$ cm

11 **a** 0.070 cm per second **b** 20.5 cm^3

12 $\frac{dV}{dt} \propto \sqrt{V} \Rightarrow \frac{dV}{dt} = -k_1\sqrt{V}$, $V \propto h \Rightarrow h = k_2V$

$$\frac{dh}{dt} = \frac{dh}{dV} \times \frac{dV}{dt} = k_2 \times (-k_1\sqrt{V}) = -k_1k_2\sqrt{\frac{h}{k_2}}$$

$$= \frac{-k_1k_2}{\sqrt{k_2}}\sqrt{h} = -k\sqrt{h}$$

13 **a** $V = \left(\frac{A}{6}\right)^{\frac{3}{2}}$ **b** $\frac{1}{4}\left(\frac{A}{6}\right)^{\frac{1}{2}}$

c $\frac{dV}{dt} = \frac{dV}{dA} \times \frac{dA}{dt} = \frac{1}{4}\left(\frac{A}{6}\right)^{\frac{1}{2}} \times 2 = \frac{1}{2}\left(V^{\frac{2}{3}}\right)^{\frac{1}{2}} = \frac{1}{2}V^{\frac{1}{3}}$

14 $V = \frac{\pi}{3}r^2h = \frac{\pi}{3}(h\tan 30°)^2h = \frac{\pi}{9}h^3$

$$\frac{dh}{dt} = \frac{dh}{dV} \times \frac{dV}{dt} = \frac{1}{\frac{dh}{dV}} \times \frac{dV}{dt} = \frac{1}{\frac{\pi}{3}h^2} \times (-6) = -\frac{18}{\pi h^2}$$

So $\frac{dh}{dt} \propto \frac{1}{h^2}$